Data-driven modelling (1)

Experimental vs observational data and their relevance to models

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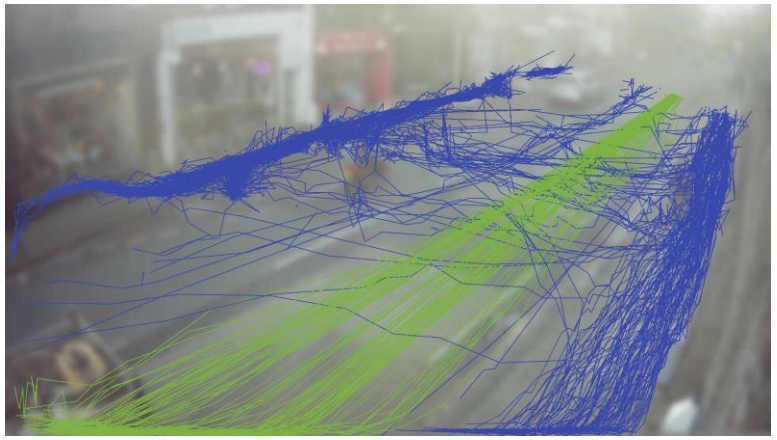
Intro to this part of the course

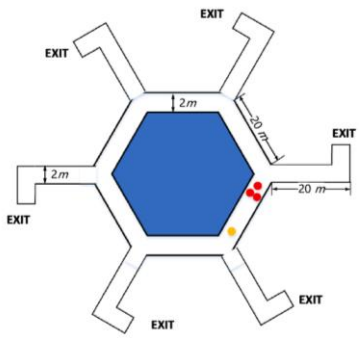
- Concerned with transport and mobility models that directly use and rely on data.

- We will cover four topics:
  1. An introduction to data (collection, issues, legal/ethical considerations)
  2. Examples for macroscopic data-driven models
  3. Examples for microscopic and mesoscopic data-driven models
  4. Time-series forecasting (short-term traffic predictions)

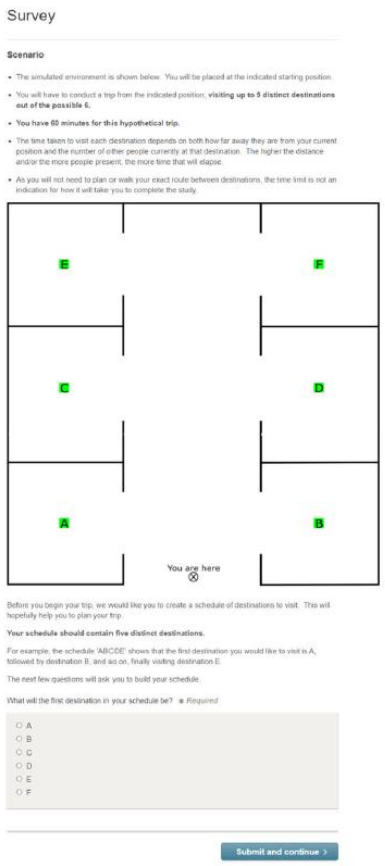
- Some approaches are alternatives to other models on the course, others are complementary.

Introductory examples

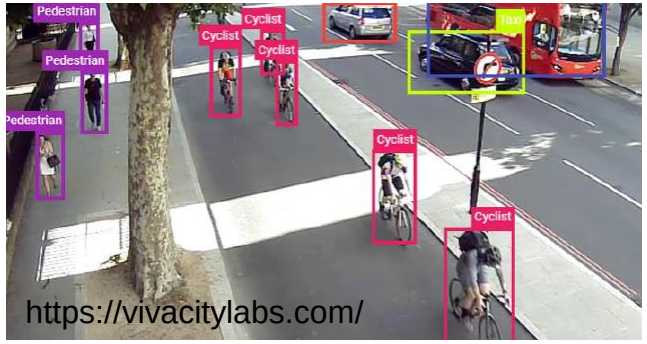


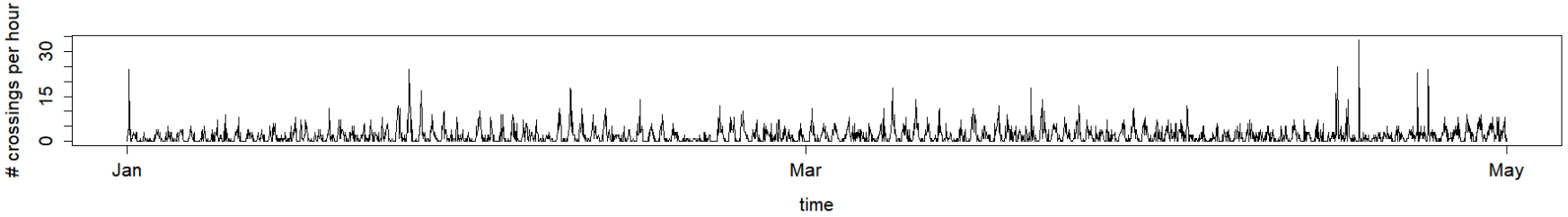












Experiments vs observational data

Experiments

- Completed under controlled conditions (e.g., lab, test track)
- Reproducible
- Often test the effect of manipulating conditions ('treatments')
- *Example: data from a driving simulator*

Observations

- Obtained from the real world, not under controlled conditions (e.g., a street, peoples' mobile phones)
- Not reproducible (typically)
- *Example: data from and induction loop*

- Distinction can be blurred, e.g., *natural experiments*
- Both experiments and observations can be done systematically

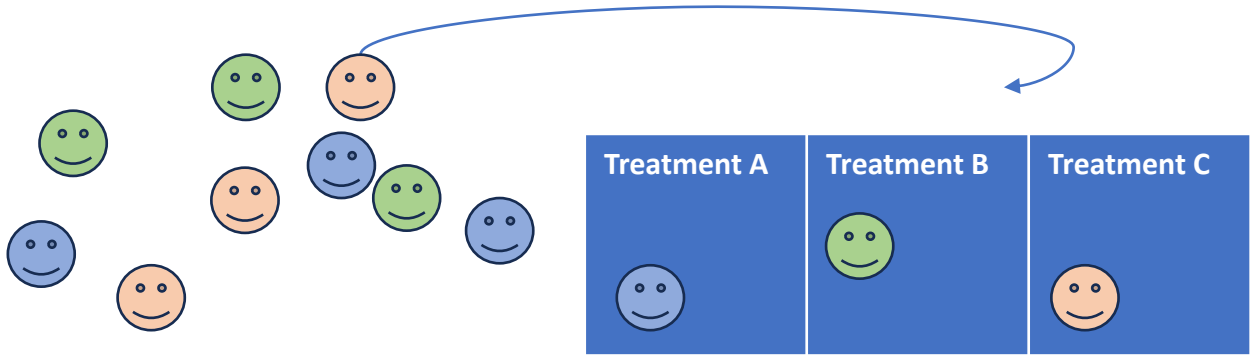
Different uses for data

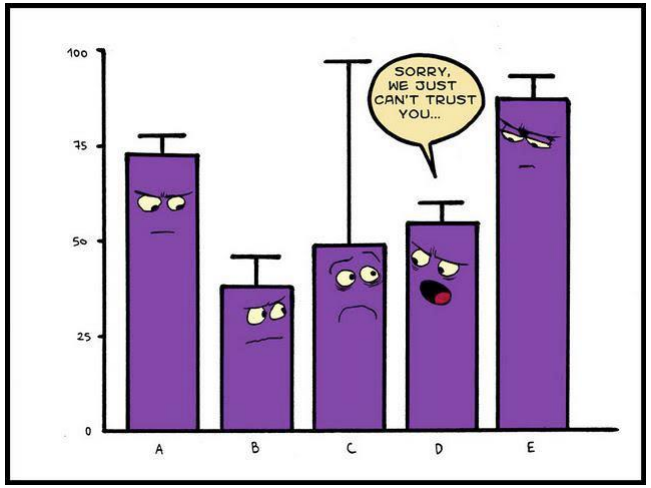
- Inference: learn something new about a process, test a hypothesis (e.g., is route length or trip duration more important in route choice?)
- Model fitting: determine the values of model parameters based on data (often a necessary step if model is to be used in applications).
- Model validation: obtain qualitative or quantitative variables that can be compared to model output (e.g., lane formation, fundamental diagram).
- Predictions: use observations of the state of a system to predict future states of the system.

- **Note:** these categories are not mutually exclusive! E.g., model selection based on the best fit can provide important insights (e.g. [1]).

Data collection through experiments

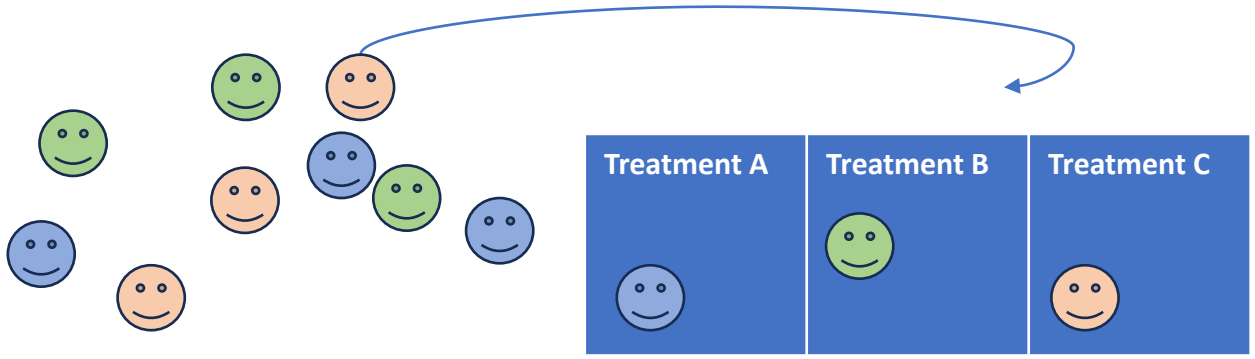
- Experimental design:
  - What's the question and how will it be measured?
  - What will we measure on (*experimental unit*)?
  - What will we vary to answer our question (*experimental conditions/treatments*)?
  - How many data points should we collect (*sample size, power analysis*)?
  - How to assign experimental units to treatments?





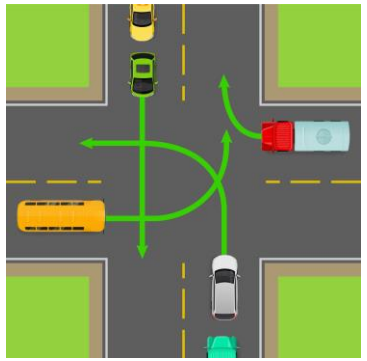
Data collection through experiments - example

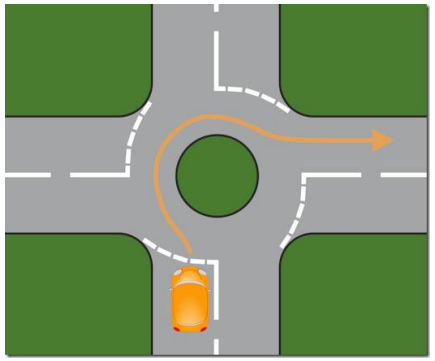
- Experimental design:
  - What's the question and how will it be measured? **Drink-driving, #pedestrians rammed**
  - What will we measure on (*experimental unit*)? **Individual drivers in simulator**
  - What will we vary to answer our question (*experimental conditions/treatments*)? **Two treatments: 1. drink one pint of water, 2. drink one pint of red wine**
  - How many data points should we collect (*sample size, power analysis*)? **1,000**
  - How to assign experimental units to treatments? **randomly**



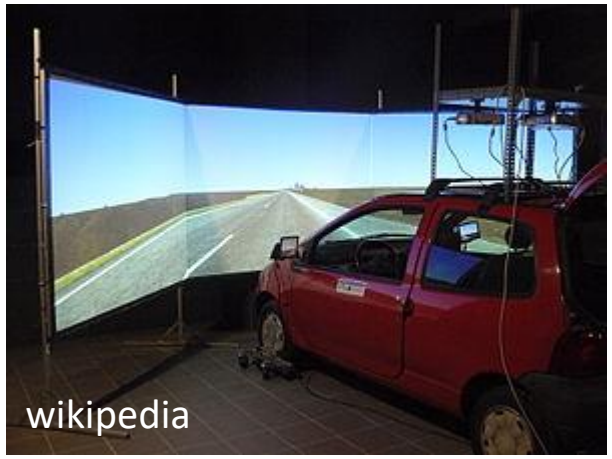
Data collection through experiments - example

- <u>Experimental design:</u>
  - What's the question and how will it be measured? **Flow through 4-way junction, vehicles/s**
  - What will we measure on (*experimental unit*)? **Entire junction on test track, inflow controlled**
  - What will we vary to answer our question (*experimental conditions/treatments*)? **Three treatments: 1. traffic lights, 2. roundabout, 3. no rules**
  - How many data points should we collect (*sample size, power analysis*)? **10 repetitions for each treatment with 100 drivers in identical cars for each experimental run**
  - How to assign experimental units to treatments? **Balanced design. Drivers are also balanced according to demographics and driving skills.**

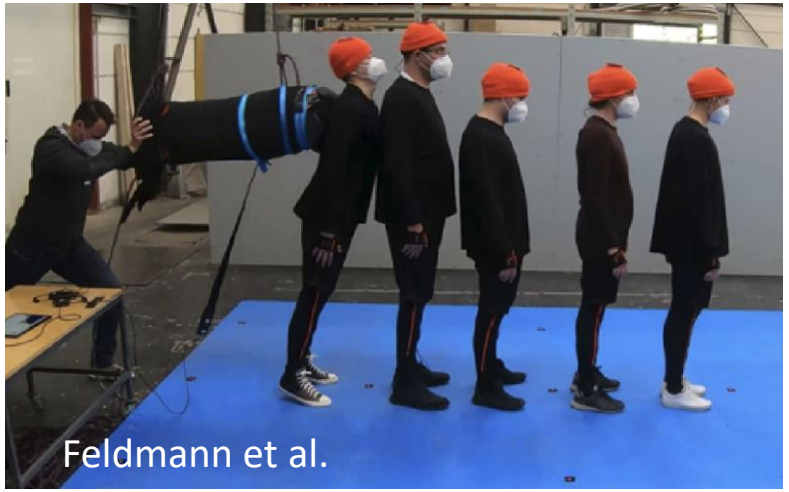


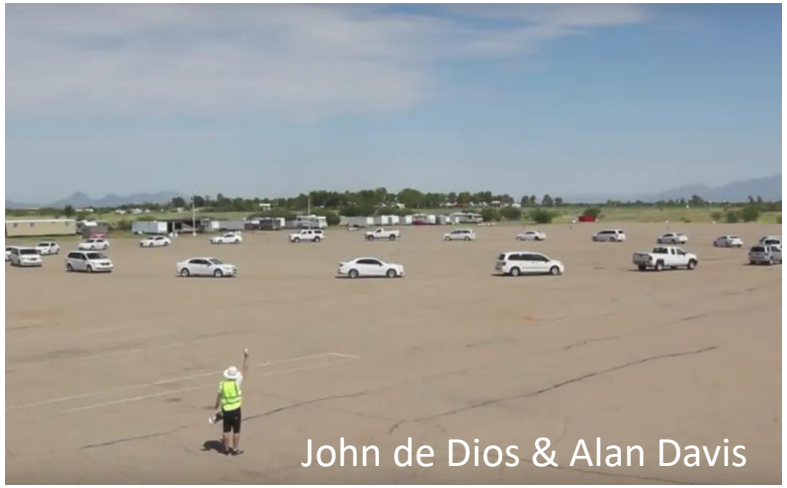


Examples for experiments











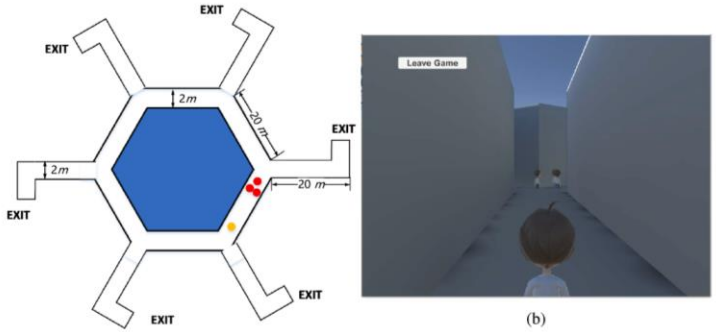
- Question: can a fire drill be an experiment?

Issues with experimental data

- Expense (time, money)



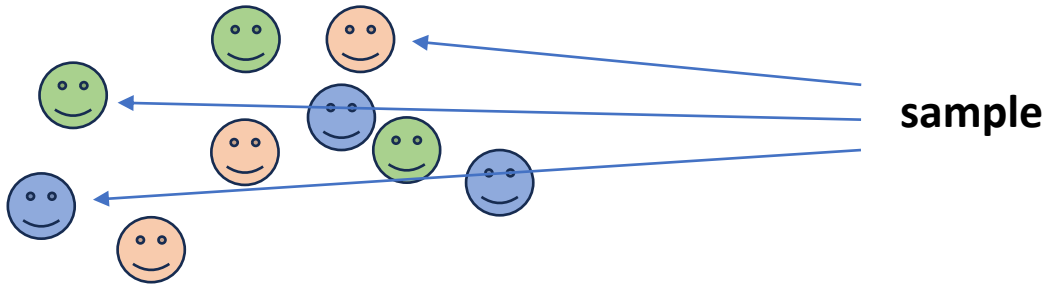
- Ecological validity
  - Do we capture real behaviour (*priming*)?
  - Limited scope?



- Sample size and reliability of results on human behaviour (consider the *crisis of reproducibility*, e.g. [2,3])

Observational data collection

- Systematic observational data collection:
  - What's the question and how will it be measured?
  - What will we measure on?
  - How many data points should we collect (*sample size*)?
  - How are data points selected for inclusion in study data (*sampling method*)?
    - Probability sampling (include items according to a probability)
    - Nonprobability sampling (systematic, e.g. rule-based sampling)



Observational data collection

- Systematic observational data collection:
  - What's the question and how will it be measured? **Continuous traffic monitoring**
  - What will we measure on? **induction loop**
  - How many data points should we collect (*sample size*)? **… number of induction loops?**
  - How are data points selected for inclusion in study data (*sampling method*)?
    All vehicles are detected (measurement error?), how is loop location selected?



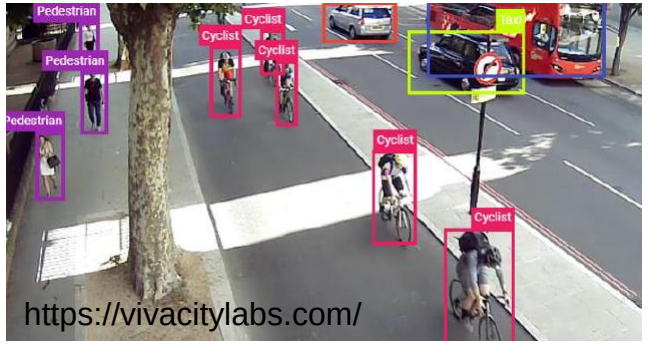
Observational data collection

- Systematic observational data collection:
  - What's the question and how will it be measured? **Mode choice of public transport users**
  - What will we measure on? **individuals**
  - How many data points should we collect (*sample size*)? **1,000 (representative?)**
  - How are data points selected for inclusion in study data (*sampling method*)?
    Every 10th passenger entering an underground station.

  - …there are obviously many variations on this theme

Examples for observational data







census
2021



Issues with observational data

- Correlation does not imply causation (see, e.g. [4]). Correlations between observed quantities does not imply they are linked (e.g., congestion charge and changes in road traffic).

- Sampling issues
  - Sample not representative (*sampling biases*, e.g. many pedestrian traffic surveys are conducted around planned developments – if used for other purposes, these data may not be representative)
  - Regression to the mean (e.g. consider placing speed cameras on roads with highest accident numbers – by chance, extreme values are expected and may be followed by values closer to the mean, possibly overestimating speed camera effectiveness)

Ethical and other considerations

- Data privacy, Data Protection Act, 2018 (UK implementation of GDPR), https://www.gov.uk/data-protection

Everyone responsible for using personal data has to follow strict rules called 'data protection principles'. They must make sure the information is:

- used fairly, lawfully and transparently
- used for specified, explicit purposes
- used in a way that is adequate, relevant and limited to only what is necessary
- accurate and, where necessary, kept up to date
- kept for no longer than is necessary
- handled in a way that ensures appropriate security, including protection against unlawful or unauthorised processing, access, loss, destruction or damage

There is stronger legal protection for more sensitive information, such as:

- race
- ethnic background
- political opinions
- religious beliefs
- trade union membership
- genetics
- biometrics (where used for identification)
- health
- sex life or orientation

There are separate safeguards for personal data relating to criminal convictions and offences.

- Research ethics
  - Declaration of Helsinki (1964) - cornerstone document on human research ethics
  - In the UK, academic research involving human participants needs to have ethics approval
  - Data privacy, plus *informed consent, right to withdraw, avoid deception, minimise harm*

Summary

- Different uses for data (inference, model fitting, model validation, predictions)
- Experimental data (principle, examples, issues)
- Observational data (principle, examples, issues)
- Legal and ethical principles

Further reading

1. Bode, et al. (2019). The emergence of macroscopic interactions between intersecting pedestrian streams. Transportation research part B: methodological, 119, 197-210.
2. Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. Science, 349(6251)
3. Fanelli, D. (2018). Is science really facing a reproducibility crisis, and do we need it to?. Proceedings of the National Academy of Sciences, 115(11), 2628-2631.
4. Stigler, S. M. (2005). Correlation and causation: A comment. Perspectives in Biology and Medicine, 48(1), 88-S94.